KUSHAL SHAH | M.S in AE/ME | PMP

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Online Portfolio – www.kushalshahmae.weebly.com | www.linkedin.com/in/kushalmae (Detailed Experience)

TECHNICAL EXPERIENCE

Technical Program Manager / Section Manager for Algorithms Section | Northrop Grumman (Azusa) ***June 2015 – Present***

Technical Program Manager [System Performance - Integrated Product Team Lead (IPTL)] (OPIR Constellation SBIRS) Jan 2023 – Present

- Led a $5M annual budget and 25-engineer team across six U.S. Space Force programs supporting OPIR payloads constellation, ensuring EVMS compliance and delivery of cost, schedule, and technical milestones. Oversaw LOS performance (pointing, control systems, Kalman filtering), radiometry (image processing, centroiding, calibrations), and mission performance testing for tactical parameter validation. Provided off-nominal support for anomaly resolution and maintained deep understanding of end-to-end payload architecture. Collaborated with PMO, Chief Engineer, Lockheed Martin, and Space Force stakeholders.
- Led as developer, product owner, and scrum master to architect full-stack apps for satellite anomaly monitoring, automated performance reporting, and ground software build analysis using Python, Flask, Streamlit, React, and SQLite—saving $1M+. Drove CI/CD and Agile practices using Atlassian tool suite to accelerate automation and deployment.
- Participated in the Architect Apprenticeship Program (AAP), dissecting customer mission requirements and conducting trade studies to architect the **Counter-Unmanned Aircraft System (C-UAS) solution, integrating radar, EO/IR systems, and AI-driven data fusion.**
- Authored technical Basis of Estimates (BOEs) for $10M+ proposals, translating system requirements into cost models, architecture diagrams, and risk profiles.

Section Manager for Mission Algorithms Section Jan 2022 – Present

- Supervised a team of 14 algorithm engineers working on image processing pipelines, tracking systems, and telemetry data analysis for space-based sensors, **overseeing $2.5M in responsibilities**. Oversaw algorithm validation in HIL/SIL test environments.
- **Directed resource planning and optimized staffing strategies** using dynamic dashboards for SEIT Director (200+).

System Performance Lead for Line of Sight (LOS) team for SBIRS Payloads April 2020 – Dec 2022

- Led and mentored a 25+ engineer team on OPIR Constellation (SBIRS), driving test design, test execution, and data analysis for LOS and Radiometry performance. Oversaw RVR closure and led root cause analysis of critical anomalies across TVAC, Launch Vehicle Integration, EOT, and sustainment phases
- Served as **Payload Technical Lead** and engineering focal point, coordinating across multidisciplinary teams—LOS, Radiometry, Thermal, GNC, Electrical, State of Health, Flight/Ground Software, Mission Planning, and Ops—during on-orbit testing and calibration for GEO5 and GEO6 launch, ensuring mission success in a fast-paced, high-stakes environment.

Responsible Engineer for LOS System Performance for GEO 5 Payload May 2018 – March 2020

- Performed system-level analysis to compute Line-of-Sight (LOS) knowledge and pointing performance, integrating data from infrared sensors, motor torque profiles, gimbal axis motion, and gyroscopes
- Developed MATLAB-based tools to analyze satellite pointing stability, Kalman filter performance, and jitter profiles—improving targeting accuracy and payload calibration.

Pathway Rotation Program – Systems Engineering & Integration (SEIT) June 2015 – May 2018

- **Rotation 1 and 3:** Led on-orbit anomaly investigations as the technical data analyst, collaborating with subject matter experts across subsystems. Served as Responsible Engineer for 24/7 Early On-Orbit Testing (EOT) campaigns on three payloads—designed test plans, executed real-time telemetry analysis, and delivered radiometric performance reports.
- **Rotation 2:** Created a MATLAB GUI under the guidance of an NG Fellow to generate missile trajectories with minimal input parameters and designed data visualizations to monitor and understand existing and emerging threats.

Thales Avionics Inc. - Research & Development Department – Mechanical Engineering Intern in Innovation Lab ***Apr 2013 – Sep 2014***

- Key member of a team of 7 engineers tasked with building prototypes and testing apertures to develop a completely immersive seat

EDUCATION

M.S in Mechanical Engineering (Control Systems) | University of California, Los Angeles (UCLA) | GPA: 3.93 June 2015

B.S in Mechanical Engineering and Aerospace Engineering *(Double Major)* |University of California, Irvine (UCI) |GPA: 3.90 June 2014

CERTIFICATES | TRAININGS

- **Project Management Professional (PMP) Certification (2024**): Earned a globally recognized certification in project management, showcasing expertise in leading projects efficiently and delivering results within scope, time, and budget.
- **SASE Top Gun 2 Leadership Training (2023):** Completed a selective program for high-potential leaders, focusing on emotional intelligence, authentic communication, personal branding, and team dynamics. Gained tools to manage stress, inspire teams, and lead with purpose**.**
- ***Caltech System Engineering Certificate (2020):*** Designed quadcopter logistic system using SysML models, architecture diagrams, and requirement traceability

TECHNICAL SKILLS

MATLAB/Simulink – Python, FLASK, Streamlit– FORTRAN – Mathematica – LABVIEW
SQL - PowerBI – Tableau– HTML / CSS – React – Node.js – Linux – LaTeX
STK – SolidWorks – NASTRAN/PATRAN – Basic GD&T/CNC Machining – 3D Printing - Root Cause Analysis
MS Office Products (Standard, Visio and Project)– Atlassian Suite (Jira, Confluence, Bitbucket, Jenkins) – Google Cloud – Azure DevOps (CI/CD)